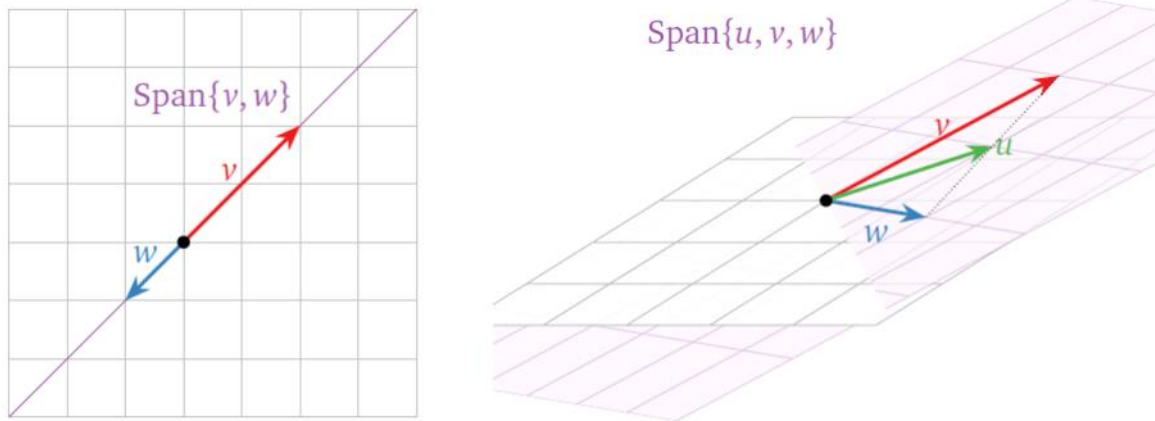


Linear Dependence, Independence, Bases & Dimension

Saturday, 14 February 2026 11:07 am



Pictures of sets of vectors that are linearly dependent. Note that in each case, one vector is in the span of the others—so it doesn't make the span bigger.

Linear Independence

$\{\text{EAST}, \text{NORTH}, \text{NORTH-EAST}\}$

Definition 16.1: Let V be a vector space and let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ be a set of vectors in V . Then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is **linearly independent** if the only scalars $c_1, c_2, \dots, c_p$ that satisfy the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0}$$

are the trivial scalars $c_1 = c_2 = \dots = c_p = 0$. If the set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is not linearly independent then we say that it is **linearly dependent**.

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We now describe the redundancy in a set of linear dependent vectors. If $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ are linearly dependent, it follows that there are scalars $c_1, c_2, \dots, c_p$, **at least one of which is nonzero**, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0}. \quad (\star)$$

For example, suppose that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ are linearly dependent. Then there are scalars c_1, c_2, c_3, c_4 , not all of them zero, such that equation $(\star)$ holds. Suppose, for the sake of argument, that $c_3 \neq 0$. Then,

$$\mathbf{v}_3 = -\frac{c_1}{c_3}\mathbf{v}_1 - \frac{c_2}{c_3}\mathbf{v}_2 - \frac{c_4}{c_3}\mathbf{v}_4.$$

Theorem 16.2: A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$, with $\mathbf{v}_1 \neq \mathbf{0}$, is linearly dependent if and only if some $\mathbf{v}_j$ is a linear combination of the preceding vectors $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$.

Explain why the following form linearly dependent sets of vectors. (Solve this problem by inspection.)

a. $\mathbf{u}_1 = (-1, 2, 4)$ and $\mathbf{u}_2 = (5, -10, -20)$ in R^3

b. $\mathbf{u}_1 = (3, -1)$, $\mathbf{u}_2 = (4, 5)$, $\mathbf{u}_3 = (-4, 7)$ in R^2

c. $\mathbf{p}_1 = 3 - 2x + x^2$ and $\mathbf{p}_2 = 6 - 4x + 2x^2$ in P_2

d. $A = \begin{bmatrix} -3 & 4 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -4 \\ -2 & 0 \end{bmatrix}$ in M_{22}

Example 16.3. Show that the following set of 2×2 matrices is linearly dependent:

$$\left\{ \mathbf{A}_1 = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \mathbf{A}_2 = \begin{bmatrix} -1 & 3 \\ 1 & 0 \end{bmatrix}, \mathbf{A}_3 = \begin{bmatrix} 5 & 0 \\ -2 & -3 \end{bmatrix} \right\}.$$

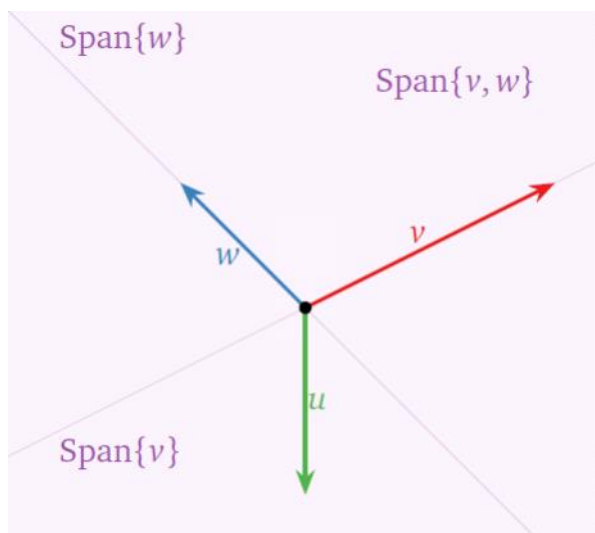
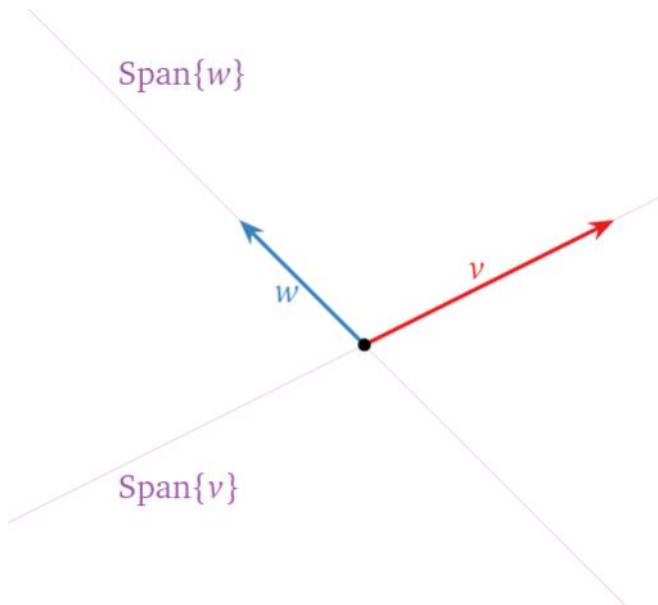
Solution. It is clear that $\mathbf{A}_1$ and $\mathbf{A}_2$ are linearly independent, i.e., $\mathbf{A}_1$ cannot be written as a scalar multiple of $\mathbf{A}_2$, and vice-versa. Since the $(2, 1)$ entry of $\mathbf{A}_1$ is zero, the only way to get the -2 in the $(2, 1)$ entry of $\mathbf{A}_3$ is to multiply $\mathbf{A}_2$ by -2 . Similarly, since the $(2, 2)$ entry of $\mathbf{A}_2$ is zero, the only way to get the -3 in the $(2, 2)$ entry of $\mathbf{A}_3$ is to multiply $\mathbf{A}_1$ by 3 . Hence, we suspect that $3\mathbf{A}_1 - 2\mathbf{A}_2 = \mathbf{A}_3$. Verify:

$$3\mathbf{A}_1 - 2\mathbf{A}_2 = \begin{bmatrix} 3 & 6 \\ 0 & -3 \end{bmatrix} - \begin{bmatrix} -2 & 6 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -2 & -3 \end{bmatrix} = \mathbf{A}_3$$

Therefore, $3\mathbf{A}_1 - 2\mathbf{A}_2 - \mathbf{A}_3 = \mathbf{0}$ and thus we have found scalars c_1, c_2, c_3 not all zero such that $c_1\mathbf{A}_1 + c_2\mathbf{A}_2 + c_3\mathbf{A}_3 = \mathbf{0}$. □

Facts about linear independence.

1. Two vectors are linearly dependent if and only if they are collinear, i.e., one is a scalar multiple of the other.
2. Any set containing the zero vector is linearly dependent.
3. If a subset of $\{v_1, v_2, \dots, v_k\}$ is linearly dependent, then $\{v_1, v_2, \dots, v_k\}$ is linearly dependent as well.



Example. The matrix

$$A = \begin{pmatrix} 1 & 2 & 0 & -1 \\ -2 & -3 & 4 & 5 \\ 2 & 4 & 0 & -2 \end{pmatrix}$$

has reduced row echelon form

$$\begin{pmatrix} 1 & 0 & -8 & -7 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Therefore, the first two columns of A are the pivot columns, so we can delete the others without changing the span:

$$\text{Span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 5 \\ -2 \end{pmatrix} \right\}.$$

Moreover, the first two columns are linearly independent.

Bases

Definition 16.4: Let W be a subspace of a vector space V . A set of vectors $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ in W is said to be a **basis** for W if

(a) the set $\mathcal{B}$ spans all of W , that is, $W = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, and

(b) the set $\mathcal{B}$ is linearly independent.

Example 16.5. Show that the standard unit vectors form a basis for $V = \mathbb{R}^3$:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Solution. Any vector $\mathbf{x} \in \mathbb{R}^3$ can be written as a linear combination of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3$$

Therefore, $\text{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \mathbb{R}^3$. The set $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is linearly independent. Indeed, if there are scalars c_1, c_2, c_3 such that

$$c_1 \mathbf{e}_1 + c_2 \mathbf{e}_2 + c_3 \mathbf{e}_3 = \mathbf{0}$$

then clearly they must all be zero, $c_1 = c_2 = c_3 = 0$. Therefore, by definition, $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis for $\mathbb{R}^3$. This basis is called the **standard basis** for $\mathbb{R}^3$. Analogous arguments hold for $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ in $\mathbb{R}^n$. $\square$

Example 16.6. Is $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ a basis for $\mathbb{R}^3$?

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -4 \\ -2 \\ 8 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 4 \\ -6 \\ -6 \end{bmatrix}$$

Solution. Form the matrix $\mathbf{A} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$ and row reduce:

$$\mathbf{A} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example 16.7. In $V = \mathbb{R}^4$, consider the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 0 \\ -2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ 2 \\ -3 \end{bmatrix}.$$

Let $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Is $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ a basis for W ?

Solution. By definition, $\mathcal{B}$ is a spanning set for W , so we need only determine if $\mathcal{B}$ is linearly independent. Form the matrix, $\mathbf{A} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$ and row reduce to obtain

$$\mathbf{A} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, $\text{rank}(\mathbf{A}) = 2$ and thus $\mathcal{B}$ is linearly dependent. Notice $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{v}_3$. Therefore, $\mathcal{B}$ is not a basis of W . $\square$

The determinant of a square matrix is zero if and only if the column vectors are linearly dependent.

Use (you must) the determinant to check whether the following set of vectors is linearly independent:

$$\left\{ \begin{bmatrix} 7 \\ -4 \\ -6 \end{bmatrix}, \begin{bmatrix} -8 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \\ -5 \end{bmatrix} \right\}$$

(**Hint:** Use these three column vectors to form a 3×3 matrix A , then compute $\det A$; if $\det A = 0$, A is not invertible, and vectors are linearly dependent; otherwise independent.)

(**Answer:** Independent)

Example 16.8. Find a basis for the vector space of 2×2 matrices.

Dimension of a Vector Space

Theorem 16.10: Let V be a vector space. Then all bases of V have the same number of vectors.

Definition 16.11: Let V be a vector space. The **dimension** of V , denoted $\dim V$, is the number of vectors in any basis of V . The dimension of the trivial vector space $V = \{\mathbf{0}\}$ is defined to be zero.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Theorem 16.12: Let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be vectors in $\mathbb{R}^n$. If $\mathcal{B}$ is linearly independent then $\mathcal{B}$ is a basis for $\mathbb{R}^n$. Or if $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \mathbb{R}^n$ then $\mathcal{B}$ is a basis for $\mathbb{R}^n$.

Example 16.13. Do the columns of the matrix $\mathbf{A}$ form a basis for $\mathbb{R}^4$?

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 3 & -2 \\ 4 & 7 & 8 & -6 \\ 0 & 0 & 1 & 0 \\ -4 & -6 & -6 & 3 \end{bmatrix}$$

Solution. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ denote the columns of $\mathbf{A}$. Since we have $n = 4$ vectors in $\mathbb{R}^n$, we need only check that they are linearly independent. Compute

$$\det \mathbf{A} = -2 \neq 0$$

Hence, $\text{rank}(\mathbf{A}) = 4$ and thus the columns of $\mathbf{A}$ are linearly independent. Therefore, the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ form a basis for $\mathbb{R}^4$. $\square$